ID - 1807190?? → $a=4$ ⇒ $A=\begin{bmatrix}2 & 1\\ 1 & 2\end{bmatrix}$ → Trace $= \lambda_1+\lambda_2 = 2+2 = 4$

$A-\lambda I = \begin{bmatrix}2 & 1\\ 1 & 2\end{bmatrix} - \begin{bmatrix}\lambda & 0\\ 0 & \lambda\end{bmatrix} = \begin{bmatrix}2-\lambda & 1\\ 1 & 2-\lambda\end{bmatrix} = (2-\lambda)^2 - 1 = 0 \rightarrow \boxed{\begin{matrix}\lambda_1=1\\ \lambda_1=3\end{matrix}} \rightarrow \lambda_1+\lambda_2 = 3+1 = 4$

For $\lambda_1=1 \rightarrow \begin{bmatrix}1 & 1\\ 1 & 1\end{bmatrix}\begin{bmatrix}x_1\\ x_1\end{bmatrix}=0 \Rightarrow x_1 = \begin{bmatrix}1\\ -1\end{bmatrix}$

For $\lambda_2=3 \rightarrow \begin{bmatrix}-1 & 1\\ 1 & -1\end{bmatrix}\begin{bmatrix}x_1\\ x_1\end{bmatrix} \Rightarrow x_2 = \begin{bmatrix}1\\ 1\end{bmatrix}$